Big Data Computing and Business Analytics

1. What role does Apache Spark typically play in Big Data analytical architectures?

a) Data storage

b) Data ingestion

c) Data processing and analytics

d) Data visualization

2. Which of the following is NOT a typical data processing framework used in Big Data analytics?

a) Apache Hadoop

b) Apache Kafka

c) Apache Hive

d) Apache Picasso

3. What is the primary benefit of using a columnar database in a Big Data analytical architecture?

a) Faster query performance

b) Lower storage costs

c) Real-time data ingestion

d) Streamlined ETL processes

4. Which component creates interactive dashboards and reports in a Big Data analytical architecture?

a) Data lake

b) Business Intelligence (BI) tools

c) Data warehouse

d) Hadoop cluster

5. In a Hadoop-based analytical architecture, what does HDFS stand for?

a) High-Density File System

b) Hadoop Distributed File System

c) Hierarchical Data File Storage

d) Hyper Data Flow Storage

6. What does OLAP stand for in the context of Big Data analytics?

a) Online Learning and Processing

b) Online Analytical Processing

c) On-Demand Load Analysis Platform

d) Offline Large-scale Analytics Protocol

7. Hadoop is often used to process and analyze large datasets. Which characteristics of Hadoop make it suitable for this task?

a) Real-time data processing and low latency hardware requirements

b) High fault tolerance and scalability

c) Strong consistency and ACID transactions

d) Small storage footprint and low

8. In Hadoop's MapReduce programming model, what is the role of the Map phase?

a) Aggregating data

b) Sorting data

c) Filtering data

d) Transforming data into key-value pairs

9. Which Hadoop ecosystem component manages and schedules resources in a Hadoop cluster?

a) HDFS (Hadoop Distributed File System)

b) YARN (Yet Another Resource Negotiator)

c) MapReduce

d) Hive

10. Which programming languages can be used to write MapReduce jobs in Hadoop?

a) Java and Python

b) C++ and Ruby

c) Scala and PHP

d) HTML and CSS

11. In Hadoop, what is the purpose of the NameNode?

a) Storing data blocks

b) Managing job scheduling

c) Storing metadata about files and directories

d) Running MapReduce tasks

12. Which Hadoop ecosystem tool is used for SQL-like querying of data stored in HDFS?

a) HBase

b) Pig

c) Hive

d) Sqoop

13. Which of the following is true regarding SparkContext's role in managing cluster resources?

a) SparkContext allocates resources at the task level.

b) SparkContext allocates resources at the executor level.

c) SparkContext allocates resources at the node level.

d) SparkContext allocates resources at the driver program level.

14.

Which SparkContext configuration parameter is used to control the level of parallelism in RDD operations?

a) spark.master

b) spark.driver.memory

c) spark.executor.cores

d) spark.default.parallelism

15. In the context of SparkContext, what is the purpose of the setCheckpointDir method?

a) To specify the location of Spark's event log.

b) To configure the level of log output.

c) To define a directory where checkpoint data will be stored.

d) To set the number of retries for failed tasks.

16. RDDs in Apache Spark are immutable. What is the primary benefit of immutability in the context of distributed data processing?

a) It allows for real-time data updates.

b) It simplifies data manipulation.

c) It ensures fault tolerance and data lineage.

d) It enables parallel processing.

17. When is data lineage information in RDDs valuable, and how does it assist in fault tolerance?

a) During data loading from external sources

b) During data transformation operations

c) During data storage in HDFS

d) During data visualization

18. In Apache Spark, what is the key advantage of using RDDs over traditional distributed storage systems like HDFS?

a) RDDs offer higher data compression.

b) RDDs allow random access to data.

c) RDDs provide in-memory data processing.

d) RDDs support complex query languages.

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20. Which of the following RDD operations is considered a transformation?

a) count()

b) reduceByKey()

c) saveAsTextFile()

d) collect()

21. RDDs in Apache Spark can be cached or persisted in memory. When might it be beneficial to use caching?

a) When dealing with small datasets

b) When there is ample storage space available

c) When an RDD is computed multiple times

d) When performing real-time data processing

22. Which of the following RDD operations triggers a full shuffle of data across the cluster?

a) map()

b) filter()

c) reduceByKey()

d) collect()

23. During which phase of the Data Analytics Lifecycle is it crucial to define the project's objectives and key performance indicators (KPIs)?

a) Data Preparation

b) Data Exploration

c) Data Modeling

d) Data Deployment

24. Which phase of the Data Analytics Lifecycle involves validating the performance of analytics models against real-world data and making necessary adjustments?

a) Data Exploration

b) Data Preparation

c) Data Modeling

d) Data Deployment

25. During which phase of a Spark job lifecycle does Spark perform optimizations like pipelining narrow transformations and selecting the best physical execution plan?

a) Data Ingestion

b) Data Exploration

c) Query Optimization

d) Data Presentation

26. In the context of Spark RDDs, what does it mean when an RDD is considered "lazy"?

a) Lazy RDDs are not processed at all.

b) Lazy RDDs are only computed when an action is called.

c) Lazy RDDs are automatically cached in memory.

d) Lazy RDDs are not fault-tolerant.

27. How does data locality optimization in Spark RDDs contribute to better performance, and in which phase of data processing is it most relevant?

a) Data locality optimization reduces network overhead by placing tasks close to the data they need, improving performance during Data Deployment.

b) Data locality optimization speeds up data loading from external sources, enhancing performance during Data Preparation.

c) Data locality optimization enables real-time data processing, resulting in faster Data Exploration.

d) Data locality optimization enhances data compression techniques, benefiting Query Optimization.

28.

In the Big Data analytics lifecycle, what challenges might organizations encounter when handling unstructured data, and how can they address these challenges effectively?

a) Challenges include data scalability and data storage costs. They can be addressed by using cloud-based solutions.

b) Challenges include data integration and data quality. They can be addressed by implementing data governance and ETL (Extract, Transform, Load) processes

c) Challenges include data security and data privacy. They can be addressed by using open-source analytics tools.

d) Challenges include data access speed and network latency. They can be addressed by optimizing network infrastructure.

20. Which technology is commonly used for distributed storage and processing of Big Data in open-source Big Data ecosystems?

a) Oracle Database

b) MongoDB

c) Hadoop

d) Tableau

30. Frequency of occurrence of an item is called _____

a) Support

b) Confidence

c) Support Count

d) Rules

31. What techniques can be used to improve the efficiency of apriori algorithm?

a) Hash-based techniques

b) Transaction Increases

c) Sampling

d) Cleaning

32. How do you calculate Confidence (A -> B)?

a) Support(A∩B) / Support (A)

b) Support(A ∩ B) / Support (B)

c) Support(A U B) / Support (A)

d) Support(A U B) / Support (B)

33. An itemset whose support is greater than or equal to a minimum support threshold is _____

a) Itemset

b) Frequent Itemset

c) Infrequent items

d) Threshold values

34. What is KDD in data mining?

a) Knowledge Discovery Database

b) Knowledge Discovery Data

c) Knowledge Data definition

d) Knowledge data house

35. In _____, the value of an attribute is examined as it varies over time

a) Regression.

b) Time series analysis.

c) Sequence discovery.

d) Prediction

36. _____ function is responsible for consolidating the results produced by each of the Map() functions/tasks.

a) Reduce

b) Map

c) Reducer

d) All of the mentioned

37. _____ part of the MapReduce is responsible for processing one or more chunks of data and producing the output results.

a) Maptask

b) Mapper

c) Task execution

d) All of the mentioned

38. _____ maps input key/value pairs to a set of intermediate key/value pairs.

a) Mapper

b) Reducer

c) Both Mapper and Reducer

d) None of the mentioned